Bishal Roy

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Applied AI/ML Engineer specializing in production **Generative AI** — multi-agent systems, Retrieval-Augmented Generation (RAG), and LLM fine-tuning. Ship citation-grounded, low-latency agentic pipelines (FastAPI, LangChain, BM25/FAISS) with measurable accuracy and latency gains. National-level hackathon record (Smart India Hackathon AIR-2) and two research papers under review.

Technical Skills

GenAI & LLMs: Multi-Agent Systems, RAG (BM25 / FAISS / ChromaDB), LangChain, LoRA / QLoRA Fine-tuning, Prompt Engineering, HuggingFace Transformers, Groq
ML / DL: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, Ensemble Methods
MLOps & Deployment: FastAPI, Docker, Git, CI/CD, MLflow, Weights & Biases, Streamlit
Data Engineering: Pandas, NumPy, Feature Engineering, Statistical Analysis, PySpark (basic)
Languages: Python, SQL

Professional Experience

Interview Kickstart — Apr 2026 – Present
Associate AI Product Manager Intern — *Remote*

- Built a GenAI evaluation pipeline that scores live-class recordings and transcripts against session ratings, using LLM-based scoring to auto-flag the 15% of sessions rating below 4.5/5 and route root-cause feedback through human-in-the-loop review — lifting average instructor rating from 4.2 to 4.6 in 8 weeks.
- Automated the weekly instructor-ops workflow in Python (auto-generating session invites, tracking availability across 40+ instructors), cutting manual scheduling effort ∼87% (∼6 hrs/week).
- Architected an AI/ML curriculum (12 modules, 4 specialization tracks) aligned to current GenAI/ML hiring bars for 150+ learners per cohort.

CloudCredits — Jul 2025 – Oct 2025
AI Engineer Intern — *Remote*

- Built and deployed financial risk-assessment models (XGBoost, Random Forest) on 5M+ transaction records, achieving 89% accuracy and reducing false positives by 34% in production.
- Engineered an automated feature-extraction pipeline over 3.2M customer records, deriving 12 domain-specific risk indicators (+27% credit-scoring precision, −18% default-prediction error).

AICTE — Nov 2024 – Dec 2024
AI Engineer Intern — *Remote*

- Fine-tuned a 7B-parameter open-source LLM (LLaMA-2) with LoRA adapters on 50K domain-specific Q&A pairs, improving answer accuracy from 72% to 91% at sub-2s inference latency.
- Delivered 8 technical workshops on GenAI, LLMs, and RAG to 600+ faculty across 15 institutions; 12 institutions deployed the curriculum within 3 months.

Technical Projects

C-TRUST: Clinical Trial Risk Intelligence System — Novartis NEST 2.0 — National Semifinalist (Top 30)
Python, FastAPI, Multi-Agent AI

- Architected a multi-agent system of 7 specialized agents with weighted consensus voting (Safety Agent at 3.0x weight), analyzing 23 concurrent clinical studies at ∼300ms latency for data-quality intelligence.
- Built a Guardian meta-agent for cross-agent consistency monitoring; added multi-layer caching, comprehensive error handling, and 331 property-based tests (Hypothesis); deployed a RESTful API via FastAPI with OpenAPI docs.

Kakehashi: Immigration-Guidance Agent System — Far Away Hackathon (International)
Python, FastAPI, BM25 RAG, Groq LLM, Next.js

- Designed a 6-agent system guiding Indian workers through Japan's immigration process, reasoning over official government sources (SSW, MOFA) with citation-backed responses — 51% accuracy vs 4% for an ungrounded baseline, with 0 hallucinations (vs 69) across 22 official SSW facts.
- Shipped production architecture: BM25 RAG over the SSW corpus, live jobs via JSearch API, Fernet AES-128 PII encryption, LLM key failover, SSE streaming, and a 22-test CI pipeline; EN/HI/JA support with PDF dossier export.

VayuNetra: Urban Air Quality Intelligence — ET AI Hackathon 2026
Python, Machine Learning, Geospatial, Groq LLM

- Built an end-to-end air-quality platform over 900+ stations across 6 Indian cities, delivering 24/48/72h PM2.5 forecasts (HistGradientBoostingRegressor) that beat persistence baselines by 27–50% on RMSE.
- Designed a multi-signal source-attribution engine (traffic, stubble burning, industrial) with ward-level confidence scores and Groq-LLM multilingual health advisories; validated with 14 passing tests on Open-Meteo CAMS data.

Additional Competition Projects

- **Cricket Scouting Intelligence** — *Rajasthan Royals SuperRR Hackathon, Rank 4 / 7,599*: engineered a Cricket Likelihood Engine over 602,992 ball-by-ball records (20+ phase-specific features), with 9 K-Means archetypes validated via LightGBM (AUC 0.614, 5-fold CV).
- **Smart Product Attribute Extraction** — *Amazon ML Challenge 2025, Top 8% / 10,000+ teams*: built a multi-modal pipeline fusing TF-IDF (SVD) text + EfficientNetB0 image embeddings with pseudo-labeling, reducing SMAPE from 44.9% to 38.4%.

Achievements & Research

- **Smart India Hackathon 2024 — All India Rank 2 (Runner-Up)** among 49,000+ teams: Mudra, an AI-powered Indian Sign Language platform (TensorFlow + MediaPipe), Ministry of Justice, Software Category.
- **Research (Under Review):** First author — "Predicting Problematic Internet Use in Children via QWK Optimization & Multi-Modal Feature Engineering" (JAIR); Co-author — "Real-Time Indian Sign Language Translation using Deep Learning" (Pattern Recognition, Elsevier).
- **BirdCLEF 2026 (Kaggle) — Top 10% globally:** 234-species bioacoustic classification using Google Perch v2 embeddings + Bayesian prior fusion (macro ROC-AUC 0.910, ongoing).
- **GDG AI/ML Co-Lead:** delivered 8 workshops on GenAI, LLMs, and RAG to 400+ developers (92% satisfaction).

Education

Dr. D. Y. Patil Institute of Technology, Pune Nov 2022 – May 2026
B.E. in Artificial Intelligence & Data Science — CGPA: 8.99/10 (9.55/10 final semester)